higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PRINCIPLES OF COMPUTER PROGRAMMING

(First Paper)

NQF LEVEL 3

(10041023)

22 November 2019 (X-Paper)
09:00–11:00

This question paper consists of 9 pages.

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TIME: 2 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Use only BLUE or BLACK ink.
6. Write neatly and legibly.

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SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 The decimal equivalent of 10011_2 is …

A 20.
B 19.
C 37.
D 21.

1.2 In Visual Basic, which one of the following represents an *explicit conversion*?

A K=CInt(Num2)
B K=Total+(Num2+Num3)
C K=400+(Num1/2)
D K=(Num1+Num2)/(Num1-Tot1)

1.3 Which one of the following is best suited to provide temporary storage for a group of elements of the same data type?

A Text-file
B Database
C Array
D Variable

1.4 The Agile software methodology used to reduce the time it takes to develop software is called …

A quick application development.
B visual application development.
C rapid application development.
D fast application development.

1.5 In Visual Basic, the statement *Dim Fruit(0) As String,* will create an array that can store … elements.

A 2
B 1
C 0
D 4

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1.6 The efficiency of a relational database system depends on the type of relationships between the various interrelated data tables. One such relationship to AVOID is the …

A one-to-one relationship.
B many-to-one relationship.
C many-to-many relationship.
D one-to-many relationship.



1.7 Compiler error messages are often used by programmers to debug the source code. Which one of the following errors will NOT produce a compiler error message?

A Semantic error
B Syntax error
C Logical error
D Runtime error

1.8 A/An … is an example of an unstructured data type.

A text-file
B variable
C relational database
D SQL



1.9 A … is an example of a post-test loop structure.

A Do...While Loop
B Do...Loop Until
C For...Next Loop
D Foreach…Loop



1.10 The significand of the floating point number 2.2596858×10^{-2} is …

A 2.2596858.
B $10^{-2.}$
C 2.
D -2.



(10 × 1) **[10]**

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.1–2.10) in the ANSWER BOOK.

2.1 An integer is an example of a string data type.

2.2 Multi-vibrators are sequential logic circuits that operate continuously between two distinct states of high and low.

2.3 One of the disadvantages of an array is that the number of elements that it will store must be known in advance.

2.4 Assigning a value to the wrong variable may result in a logical error.

2.5 It is possible to search the content of text files using SQL.

2.6 Data validation is the process of checking that your program has no syntax errors.

2.7 To calculate 2's complement, you have to divide the number by 1011_2.

2.8 Floating point numbers are represented as binary integers inside a computer.

2.9 Input Masks are used to restrict how and what data the user enters.

2.10 The primary key of a relational database table uniquely identifies the database.

(10 × 1) **[10]**

QUESTION 3

Complete the following sentences by filling in the missing word or words. Write only the word or words next to the question number (3.1–3.5) in the ANSWER BOOK.

3.1 A/An … is a step-by-step approach to solving problems.

3.2 A/An … number is a base 8 number.

3.3 A/An … is used for permanent storage of structured data.

3.4 *Dim intNumber(4,5) As Integer* is an example of a/an … array declaration.

3.5 Logical variables are also known as … variables.

(5 × 1) **[5]**

TOTAL SECTION A: 25

SECTION B

QUESTION 4

4.1 Find 2's complement of 10011011. Show ALL steps. (4)

4.2 Draw a logic diagram of the following: A and B OR C and NOT D (4)

4.3 Draw the symbol for a stable multi-vibrator. (2)
[10]

QUESTION 5

5.1 If A = -2, B = 10, C = 3, D = TRUE and E = FALSE, evaluate the following expression:

A < 5 and B = 10 or NOT D and C > A or NOT E. Show ALL steps. (5)

5.2 Define each of the following concepts:

5.2.1 Constant

5.2.2 Variable

5.2.3 Modular programming

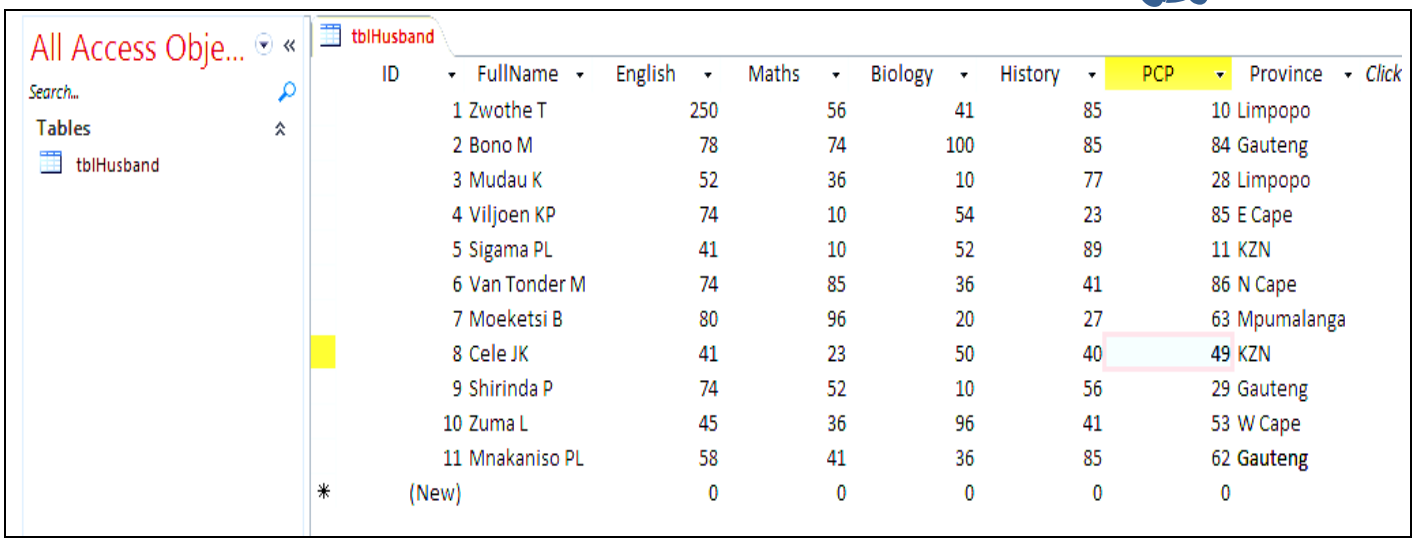
5.2.4 Event-driven programming

5.2.5 Database record

(5 × 1) (5)
[10]

QUESTION 6

Study the database screenshot below and answer the questions. The name of the table is tblHusband.



All Access Obje...
Search...
Tables
tblHusband

tblHusband

ID	FullName	English	Maths	Biology	History	PCP	Province
1	Zwothe T	250	56	41	85	10	Limpopo
2	Bono M	78	74	100	85	84	Gauteng
3	Mudau K	52	36	10	77	28	Limpopo
4	Viljoen KP	74	10	54	23	85	E Cape
5	Sigama PL	41	10	52	89	11	KZN
6	Van Tonder M	74	85	36	41	86	N Cape
7	Moeketsi B	80	96	20	27	63	Mpumalanga
8	Cele JK	41	23	50	40	49	KZN
9	Shirinda P	74	52	10	56	29	Gauteng
10	Zuma L	45	36	96	41	53	W Cape
11	Mnakaniso PL	58	41	36	85	62	Gauteng
*	(New)	0	0	0	0	0	

6.1 Write SQL queries for the data requests below.

6.1.1 Show ALL the columns for students from Limpopo. (3)

6.1.2 Show ALL the columns, including the TOTAL mark obtained for each student. (5)

6.2 Explain the function of each of the following aggregate functions:

6.2.1 Sum

6.2.2 Min

6.2.3 Max

(3 × 1) (3)

[11]

QUESTION 7

7.1 List THREE permanent storage data structures and give one feature of each. (3 × 2) (6)

7.2 Examine the following data and answer the questions.

25	**50**	**2**	**77**	**90**	**41**	**21**	**74**	**85**	**40**

7.2.1 List the steps, in the correct sequence, that a SELECTION SORT will follow to sort the items in ascending order. (4)

7.2.2 Name TWO search algorithms that can be used to search a data structure. (2)
[12]

QUESTION 8

8.1 Name THREE possible causes of input errors. (3)

8.2 Explain the use of *checksum* in error detection. (2)

8.3 Study each of the following statements and identify the type of error:

8.3.1 The program runs correctly, but the output is wrong.

8.3.2 The compiler indicates variable not declared.

8.3.3 The improper or incorrect use of programming statements in the application.

(3 × 2) (6)
[11]

QUESTION 9

Study the following scenario and answer the questions.

As a programmer, the college requested you to design a program that will accept the age of a person and determine whether or not he/she is allowed to vote. The output must be in the form a message, which should be assigned to a string variable called *strMessage*.
The message can either be 'You are allowed to vote' or 'You are NOT allowed to vote'.
The program must only allow a person from 18 or above to vote.

9.1 Analyse the program processes by developing an IPO chart. (5)

9.2 Study the following VB.Net code snippet and answer the questions.

```
For i As Integer = 10 To 1 Step 1
    lstDisplay.Items.Add("I am tired ")
Next
```

9.2.1 What will be the output of this piece of code?

9.2.2 What, if anything, is wrong with this piece of code?

9.2.3 Identify a method, if any, in this piece of code.

(3 × 2) (6)

[11]

QUESTION 10

Study the following scenario and answer the questions.

As a programmer, you were taught to practise the principles of defensive programming. You learned that the programs you write will be used by different users with varying backgrounds. Some of these users are refered to as 'thoughtless users'.

10.1 10.1.1 Define what is meant by a '*thoughtless user*'. (2)

10.1.2 State FIVE defensive programming techniques that you will apply in your program to sidestep or bypass the actions of a 'thoughtless user'. (5)

10.2 Name any THREE data validation methods you can use in your code to get around the actions of a 'thoughtless user'. (3)

[10]

TOTAL SECTION B: 75
GRAND TOTAL: 100

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